

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:05:35.372140
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

- 1 Isolate the damaged area**
Clearly mark and isolate the high-severity dent on the aircraft's surface to define the repair scope.
- 2 Remove the dent**
Carefully remove the dent using appropriate tools, ensuring not to introduce additional stresses or damage to the surrounding metal.
- 3 Prepare the surface**
Clean and prepare the dented area, removing any debris or corrosion to ensure a good repair surface.
- 4 Apply the repair material**
Select and apply the same type and thickness of material used in the original construction to the dented area, ensuring a seamless integration.
- 5 Cure and inspect the repair**
Allow the repair material to cure as per the manufacturer's instructions, then inspect the repair to ensure it meets the required strength and aesthetic standards without altering the aircraft's resonant frequency.